



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering

Academic Year 2025 – 2026 (Odd Semester)

Degree, Semester & Branch: VII Semester B.E. CSE 'A' & 'B'

Course Code & Title: CCS342 DevOps

Name of the Faculty member (s): Dr.B.Vijayalakshmi, ASP-II/CSE

Innovative Practice Description

- **Unit / Topic:** Unit I / Introduction to AWS, GCP, Azure
- **Course Outcome:** CO1
- **Topic Learning Outcome:** 1b
- **Activity Chosen:** Real World Learning – Cloud Platform Discovery challenge

- **Justification:**

The students have the exposure of different cloud services in previous subjects. Hence there is a need to have explore more about these cloud services with respect to DevOps functionalities. This activity transforms learners into *knowledge discoverers* by assigning each team a cloud platform (GCP, AWS, Azure) and asking them to map DevOps functions (CI/CD, IaC, Monitoring, etc.) to real services. Instead of memorising tools, students analyze, compare, and justify platform choices using authentic DevOps workflows.

Time Allotted for the Activity: 40 Minutes

- **Details of the Implementation:**

- The students were divided into three teams, with each team assigned one cloud platform: AWS, Azure, or Google Cloud Platform (GCP).
- Each team was given a structured comparison template containing two sections:
 - DevOps Function Mapping – Teams had to identify the corresponding cloud service for various DevOps functions such as:
 - Source Code Repository, CI/CD Pipeline, Infrastructure as Code (IaC), Container Orchestration, Container Registry, Monitoring & Logging, Configuration Management, Secrets Management.
 - Platform Summary Table – Students filled in key factual insights such as:
 - Platform Name, Launch Year, Owned By, Compute & Storage Services, Free Tier Options, Example Industry Use, and One Unique Feature.
- The teams conducted guided research using official cloud documentation and community resources and completed the features of the cloud service for the time duration of 25 minutes and it is shown in Fig 1.
- A collaborative discussion session was held where each team presented their findings for the duration of 5 minutes and answered cross-questions from peers.

- The team members Harini P, Karthiga R, Nithya R, Krishnaveni M presented the findings about GCP cloud services and it is shown in Fig 2.
- The cross-team presentations enable *experiential learning*, turning cloud fundamentals into practical decision-making skills—exactly what industry expects from DevOps engineers.
- A comparison board was created to consolidate all findings, enabling reflection and peer evaluation.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1	3	2	3	3	2	1	1	2	3	1	1	3	2	2

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO2	PO4	PO5	PO9	PO10	PO12	PSO2
	2	2	2	1	1	1	3
Justification for correlation	Students compare multiple cloud platforms and choose appropriate DevOps services for real-world use cases.	Students explore official documentation and industry sources to map DevOps services across platforms.	Students discover cloud-native DevOps tools like Terraform, Jenkins, Kubernetes, IAM, etc.	The students function effectively as a team.	Teams present findings and justify choices during peer discussions.	Engage in independent and continuous learning	The idea of different devops cloud tools are essential to developing IT solutions.

• Images / Screenshot of the practice:

Karthika P
Krishnaveni
Nithya R
Harini P

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Cloud Platform Discovery Challenge

DevOps Function	Cloud Service Name	Description
Source Code Repository	Cloud Source Repositories	Google Managed private Git repo (GCP IAM + CI)
CI/CD Pipeline	Cloud Build, cloud deploy	Use serverless CI/CD pipeline to build, test & deploy
Infrastructure as Code (IaC)	Terraform, Ansible, Cloud Deployment	Automation of configuration management through Terraform
Container Orchestration	Kubernetes, Anthos	For container orchestration and auto scaling, multi/hybrid cloud Kubernetes
Container Registry	Artifact Registry	To maintain Docker images, files engine
Monitoring & Logging	Cloud Monitoring, Error logging, Cloud Audit, Opsgenie	Automated logging & monitoring
Configuration Management	Terraform, Ansible, Cloud Config	For automated config of Infra
Secrets Management	Secret Manager	Versioning, access control audit logging

Question	Answer
Platform Name	Google Cloud Platform
Launch Year	2008
Owned by	Google
Compute Service	Google Compute Engine
Storage Service	Big Table, BigQuery, Cloud SQL, Cloud Storage, Filestore
Free Tier Option	\$300 free credits
Example Industry Use	Spotify, Snapchat, Twitter, PayPal
One Unique Feature	Kubernetes for container automation

Storage: cloud storage, Block, File store

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Cloud Platform Discovery Challenge

DevOps Function	Cloud Service Name	Description
Source Code Repository	Azure Repos	Supports branching, pull and code reviews
CI/CD Pipeline	Azure Pipeline	Used for Continuous Integration and Continuous Deployment
Infrastructure as Code (IaC)	ARM template, Azure Bicep	Manage default template [Editing]
Container Orchestration	Azure Kubernetes Service	Auto scaling and self healing containers
Container Registry	ACR	Build, store, manage and deploy images, OCI artifacts
Monitoring & Logging	Azure Monitor	Used to Monitor log and activity
Configuration Management	Azure Automation State Configuration	Automatically apply and monitor system settings
Secrets Management	Azure Keyvault	Storing and accessing secure key, certificate and sensitive information

Question	Answer
Platform Name	Microsoft Azure
Launch Year	Feb 1, 2010
Owned by	Microsoft
Compute Service	Virtual Machine, Container Instance
Storage Service	Storage [Blob Storage], File Storage
Free Tier Option	12 months free credits for new customers
Example Industry Use	Amazon, Apple, NVIDIA
One Unique Feature	Deep Integration of Microsoft services

Fig. 1: Features of Cloud Service



Fig 2. Discussion Session

• Reflective Critique:

❖ Feedback of practice from students and other stakeholders:

- Students found the activity is more engaging than traditional lectures, as it allowed them to discover and compare real-world tools rather than passively receiving information.

- Many students expressed that mapping DevOps functions to cloud services helped them clearly understand why different tools exist and when to use them.
- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
 - Collaborative discussion promotes peer teaching, as each team becomes the “expert” for its platform.
 - Encouraged critical thinking, teamwork, and technical articulation, all essential for employability in cloud and DevOps roles.
- ❖ ***Challenges faced in implementation:***
 - Some students were initially overwhelmed by unfamiliar technical terminology.
 - Variations in digital literacy caused unequal contribution among team members.
 - Ensuring authentic sources of information was difficult as some students relied on outdated third-party websites instead of official documentation.

References:

- <https://teaching.cornell.edu/teaching-resources/active-collaborative-learning/collaborative-learning>

Signature of Faculty Member

HOD